Midterm Exam for "The Mathematical Foundations of Political Methodology"

You have our class time for this exam. You may use a calculator and refer to your Pemberton & Rau textbook. Please do not use the internet during the exam. Show your work and be tidy!

1. Given the following hold, what (if anything) do we know about the relationship between sets A and C?

   (a) $A \cap B = A$ and $B \cap C \neq \emptyset$ Nothing.

   (b) $A \cup B = A$ and $B \cap C \neq \emptyset$ $A \cap C \neq \emptyset$.

   (c) $C \cup D = B$ and $B \cap C = A$ $A = C$.

2. Find the limit of each of the following sequences as $n \to \infty$, if the limit exists.

   (a)
   $$\frac{4n^3 - 1}{(n+1)^3}$$
   It's 4.

   (b)
   $$\frac{(-1)^n n^{\frac{5}{8}}}{7 + n^{\frac{7}{8}}}$$
   It's 0.

   (c)
   $$\frac{(-1)^n n^{\frac{1}{2}}}{n^{\frac{1}{2}} + n^{\frac{2}{5}}}$$
   It doesn't exist.

3. Use the (δ, ϵ) definition of a limit to prove that the function
   $$f(x) = 1/2x - 10$$
   is everywhere continuous.

   Proof: We need that limit as $x \to x_o$ of $f(x)$ equals $\frac{1}{2}x_o - 10$ for all x_o.

   For any $\epsilon > 0$, we can find a $\delta > 0$ such that $|\frac{1}{2}x - 10 - \frac{1}{2}x_o + 10| < \epsilon$ whenever $|x - x_o| < \delta$.

   Rearranging, we get $\frac{1}{2}|x - x_o| < \epsilon$ when $|x - x_o| < \delta$.

   Setting $\delta = 2\epsilon$ proves the result.

4. Let

$$f(x) = x^2/(x+4)$$

(a) Find the critical points of f and characterize them as maxima, minima, or inflection points.

Critical points at $x = 0, x = -8$; -8 is a local maximum, 0 is a local minimum.

(b) Find the regions on which f is increasing and decreasing.

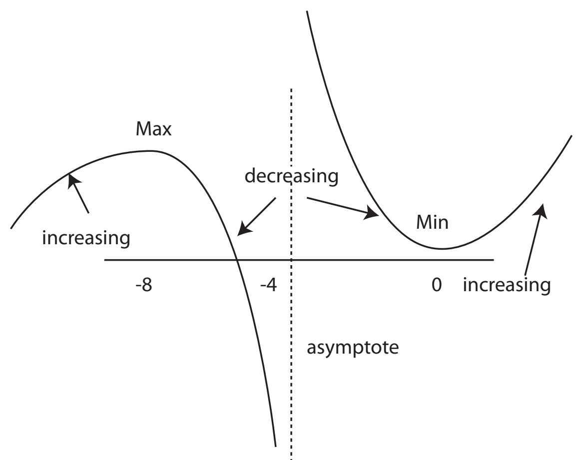
f is increasing on $(-\infty, -8)$, decreasing on $(-8, 0)$, and increasing on $(0, \infty)$.

(c) Find any asymptotes that f may have.

As $x \to -4$ from above $y \to \infty$. From below, $y \to -\infty$.

EXTRA CREDIT: As $x \to \pm\infty$ f approaches the line $y = x$.

(d) Use your answers above to plot out f.



5. Perform the following matrix operations:

(a) Find the inverse of

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

It is

$$\begin{bmatrix} 1 & -1 & -3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Find the rank of

$$\begin{bmatrix} 1 & 1 & 5 \\ 0 & 2 & 4 \\ -2 & 0 & -6 \\ 1 & 9 & 21 \end{bmatrix}$$

It is 2.

(c) Check whether the following matrix is positive definite, positive semidefinite, negative definite, negative semidefinite, or none of the above.

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

It is none of the above (the 2X2 principal minors aren't positive for A or non-negative for $-A$.).

6. Find and classify all of the critical points of the following functions:

(a)

$$f(x, y) = -10x^2 + 4xy - \frac{2y^2}{5}$$

There are a continuum of critical points: all points of the form $(C, 5C)$ with $C \in \mathbb{R}$. They are all global optima, because f is globally concave: Its Hessian is

$$\begin{bmatrix} -20 & 4 \\ 4 & -\frac{4}{5} \end{bmatrix}$$

so it has determinant 0 and its diagonals are negative. Thus the Hessian is negative semidefinite.

(b)

$$f(x, y, z) = -x^2 + \log(x) - \frac{1}{y} - y - z^2 + z \quad \text{for } x, y, z > 0$$

There is one critical point: $(\frac{1}{\sqrt{2}}, 1, \frac{1}{2})$. It is a global optimum because the function is the sum of concave functions of one variable. No Hessian required.

(c)

$$f(x, y) = x^2 + y^2 - xy \text{ subject to the constraint that } 2x - y = 3$$

The Lagrangian is

$$L(x, y, \lambda) = x^2 + y^2 - xy - \lambda(2x - y - 3).$$

The first order conditions imply that $2x - y - 2\lambda = 0$, $2y - x + \lambda = 0$, and $2x - y - 3 = 0$. Solving for x and y we get $(\frac{3}{2}, 0)$ as a critical point (incidentally, $\lambda = \frac{3}{2}$). It is a constrained (global) minimum. We know this because $x^2 + y^2 - xy$ is globally convex and our constraint is linear. The Hessian of f is

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

which has determinant $3 > 0$ and positive diagonals.